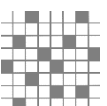


PCIe Adapter Cards

Document Revision: 1.0

User Guide

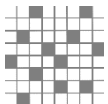


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Revision history

Revision	Description	Date
1.0	Initial Release (Preliminary)	2025-02-19

Introduction

This document lists ViALUX approved PCIe adapter cards and cables for UltraSpeed V-Modules and provides detailed instructions on installing and configuring the adapter card to ensure optimal performance.

The PCIe adapter cards enable external PCIe connectivity for UltraSpeed V-Modules with SFF-8644 cables and support both copper and active optical cable (AOC).

NOTE: Only PCIe adapter cards and cables configured and tested by ViALUX are supported.

PCIe Adapter Cards

This section lists PCIe adapter cards available from ViALUX.

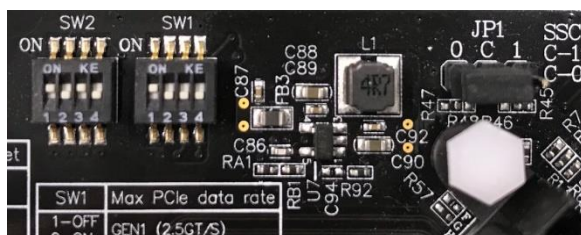
H-0127 (PCIe x8 to 2x SFF-8644)



External PCIe (2x SFF-8644) to PCIe x8 Gen3 Switch Host Card:

- Dual SFF-8644 Connector (Up to 2 UltraSpeed V-Modules)
- Low Profile Form Factor
- Requires PCIe x8 or x16 Slot

Configuration and DIP Switch Settings:



Default Configuration:



H-0130 (PCIe x16 to 4x SFF-8644)



External PCIe (4x SFF-8644) to PCIe x16 Gen3 Switch Host Card:

- Quad SFF-8644 Connector (Up to 4 UltraSpeed V-Modules)
- Low Profile Form Factor
- Requires PCIe x16 Slot

Configuration and DIP Switch Settings:



Default
Configuration:



PCIe Cables

PCIe Gen3 (SFF-8644) cables available from ViALUX:

Part	ViALUX Order Number
Copper Cable, 3 m	K-0289
Active Optical Cable, 30 m	K-0290

Installation

Step 1 - Unpack the Board

The PCIe adapter cards are shipped in an anti-static bag to prevent static electric damage. The card should only be removed from the bag if anti-static precautions are taken.

Step 2 – Check the Board Configuration

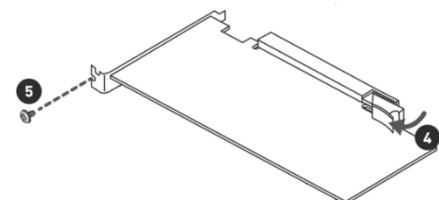
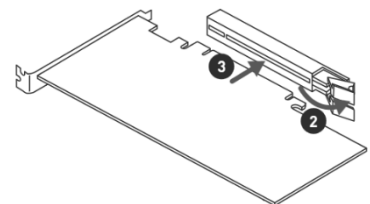
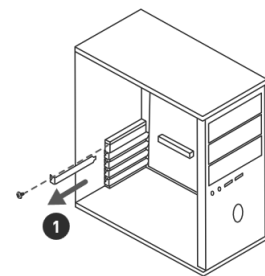
Check if the DIP switches are in default configuration. Please refer to the section *Configuration and DIP Switch Settings* of your card model.

Step 3 – Install the Adapter Card

Before installing the adapter card, make sure you are properly grounded to avoid static discharges that may destroy your computer or the adapter card. Do not install the adapter card while the PC is powered on.

To install the adapter card into a computer proceed as follows:

- **Power Down** – Shut down the computer and unplug the power cable.
- **Open the Case** – Remove the side panel of the computer. Use a screwdriver if necessary.
- **Locate a PCIe Slot** – Identify an available PCIe slot on the mainboard.
- **Remove Slot Cover** – If the slot is covered, unscrew and remove the metal slot cover from the case ❶. Keep the screw for later.
- **Align the Card** – Hold the PCIe card by its edges and align its connector with the PCIe slot. Ensure the bracket aligns with the expansion slot opening on the case. If present, ensure that retention clip at the end of the slot is released ❷.
- **Insert the Card** – Firmly but gently press the card straight down into the PCIe slot until it is fully seated ❸. The retention clip at the end of the slot, if present, should snap into place ❹.
- **Secure the Card** – Use the screw to secure the card's bracket to the case ❺.
- **Close the Case** – Attach the side panel screw it back in place.



Please refer to your mainboard or computer case manual for specific instructions.

The cards support PCIe Gen1, Gen2 and Gen3 signaling. To achieve maximum performance a PCIe Gen3 slot is required.

Step 4 – Installing and Removing the Cable

Installing and removing cables should be done with the computer powered off.

- **Connecting the Cable** – Carefully install the cable connector **1** into the connector housing **2**. Use even pressure to insert the connector until it is secure.
- **Disconnecting the Cable** – Carefully pull the release tab **3** to release the cable from the locking latches and gently pull the cable out of the connector.

